

Anurag Chatterjee Week 1 Updates: UEIL Internship

Date: 5/29/26

1. **Objective:** The main objective of this task is to differentiate between AVNRT vs AVRT from 3-channel intraoperative ECG signals. This is important because current clinical trials rely heavily on P wave and QRS characteristics which can often be unreliable by nature and hence, we need to find an accurate ML classifier to classify this.

2. **Dataset Used:**
 - Total recordings: 46 (.mat files, fs=10,000 Hz, ~7.5s each)
 - AVNRT (Class 0 In ML Model): 20 recordings across 8 patient sessions
 - AVRT (Class 1 In ML Model): 26 recordings across 7 patient sessions
 - Channels Used: Ch3 (primary as all features were statistically significant), Ch4 (2 features as they were more statistically significant), Ch1 (excluded)
 - Ch1 Exclusion: Noisy and irregular signals

3. **Final Features Used And Extracted:** Feature extraction followed a systematic pipeline across eight categories: **(1) temporal/HRV, (2) morphological, (3) interval, (4) QRS alternans, (5) spectral, (6) nonlinear/Poincaré, (7) wavelet DWT energy, and (8) R-peak amplitude statistics.**
 - *ch3_pNN50_pct*
 - *ch3_HR_std_bpm*
 - *ch3_beat_skewness_mean*
 - *ch3_beat_kurtosis_mean*
 - *ch3_QRS_duration_ms_mean*
 - *ch3_P_wave_presence_ratio*
 - *ch3_ST_slope_mean*
 - *ch3_R_peak_count*
 - *ch3_RP_interval_mean_ms*
 - *ch3_RP_PR_ratio*

- *ch3_SD2_ms*
- *ch3_SD1_ms*
- *ch3_SD1_SD2_ratio*
- *ch3_RR_std_ms*
- *ch3_RMSSD_ms*
- *ch3_RR_mean_ms*
- *ch3_P_wave_absent_flag*
- *ch4_SD1_SD2_ratio*
- *ch4_P_wave_absent_flag*

4. **Literature Review Summary:** Two parallel literature reviews were conducted to inform feature selection and establish comparison baselines.

For general SVT and cardiac arrhythmia classification context, seven papers were reviewed covering machine learning approaches to ECG-based arrhythmia detection. Key references include a gradient boosting model achieving AUC 0.97 using 40 hand-crafted ECG features (Hannun et al., PubMed 38246906), a CNN-LSTM achieving 95.63% accuracy on SVT vs VT classification (arXiv 2507.14196), a decision tree model using autoregressive features achieving 97% accuracy (arXiv 2112.12953), a VT-specific CNN using time-frequency representations achieving 99.09% (PMC10532022), and a 2025 CardioForest ensemble model (arXiv 2509.25804). These papers collectively informed the feature categories extracted in this work - ***particularly HRV temporal features, QRS morphological features, spectral band power, and Poincaré nonlinear features.***

For AVNRT vs AVRT classification specifically, five papers were reviewed. This distinction is critical because most published ML cardiac papers address the broader SVT vs VT problem, whereas AVNRT vs AVRT is a harder intra-SVT subtype discrimination problem. ***The primary comparison baseline is Paper A, a CNN trained on 12-lead ECG achieving 77.4% AUC on AVNRT vs AVRT classification (ScienceDirect,***

doi:10.1016/j.hroo.2023.01.001). Additional references include a ResNet-34 model achieving AUROC 0.726 (Paper B), and three clinical algorithm papers (Papers C–E) establishing that RP interval is the primary electrophysiological discriminator between AVNRT and AVRT - with $RP' < 90\text{ms}$ indicating AVNRT (retrograde P buried in or immediately after QRS) and $RP \geq 100\text{ms}$ indicating AVRT (visible retrograde P between QRS and T-wave). Papers D and E specifically quantified mean RP' of 86.3ms for AVNRT and $RP \geq 100\text{ms}$ as diagnostic of AVRT in paediatric populations, providing the clinical ground truth against which this work's RP interval extraction was validated.

5. **Signal Processing Pipeline:**

- a. *Step 1 - Bandpass filter: Butterworth 0.5–50 Hz (SOS form, $fs=10,000$ Hz, $order=4$, zero-phase `sosfiltfilt`)*
- b. *Step 2 - R-peak detection: `scipy find_peaks` ($height=0.5$, $distance=200\text{ms}$ / 2000 samples). Note: NeuroKit2 Pan-Tompkins bypassed - halves and beat count at 210+ BPM due to refractory period.*
- c. *Step 3 - Waveform delineation: NeuroKit2 `ecg_delineate` (DWT method, pre-supplied R-peaks). This extracts: P onset/peak/offset, Q, S, T offset*
- d. *Step 4 - Feature extraction: 8 categories, 91 features per channel extracted per recording.*

6. **Machine Learning Models Used (Logistic Regression, Random Forest And XGBoost):**

- Dataset: $n=46$ recordings (26 AVRT, 20 AVNRT), 19 features
- Training Data: 36 recordings (16 AVNRT, 20 AVRT)
- Testing Data: 10 recordings (4 AVNRT, 6 AVRT)
- Train/Test Split: 80/20
- Three binary classifiers trained On Logistic Regression, Random Forest And XGBoost (AVNRT=0, AVRT=1)
- Common Pipeline:

- Median imputation → StandardScaler (Logistic Regression only)
- Stratified 80/20 train/test split
- StratifiedKFold (5-fold) GridSearchCV for tuning
- Leave-One-Out CV (LOO-CV) as primary metric [rationale: n=46 too small for reliable hold-out]

7. Machine Learning Models Analysis:

- **Model 1: Logistic Regression** - Best params: C=10, penalty=L2, solver=saga
- **Model 2: Random Forest** - Best params: max_depth=3, n_estimators=50, min_samples_split=2, class_weight: balanced
- **Model 3: XGBoost** - Best params: learning_rate=0.1, max_depth=2, n_estimators=100

8. Results - ML Models Comparison:

Primary metric: LOO-CV accuracy (most reliable for n=46 recordings in total)

Model	LOOCV Accuracy	K Fold Cross-Validation (k-5) Accuracy	Correct Classifications
Logistic Regression	80.43%	83.57%	37/46
Random Forest	78.26%	78.57%	36/46
XGBoost	89.13%	77.86%	41/46
Paper A Reference Benchmark (CNN, 12 Led) 77.4% AUC			

9. Key Findings:

- XGBoost outperforms published 12-lead CNN baseline (89.13% LOO-CV vs 77.4% AUC) using 3-channel ECG
- pNN50, HR variability, QRS morphology (skewness, kurtosis, duration) are most discriminating features

- RP interval separates classes statistically ($d=1.06$, $p=0.001$) but has 28% missing data at 200+ BPM
- Feature importances are clinically consistent with known electrophysiology of AVNRT vs AVRT
- The majority of logistic regression coefficients are directionally consistent with clinical expectations for AVNRT vs AVRT. Features showing counterintuitive signs - specifically `RP_interval_mean_ms`, `QRS_duration_ms_mean`, and `R_peak_count` — are attributed to multicollinearity between correlated features. When correlated features are included simultaneously in a logistic regression model, individual coefficients become unreliable indicators of direction while the combined model prediction remains valid. This is a known limitation of logistic regression in high-feature-count settings relative to sample size.

10. **Limitations Compared To Literature Reviews Conducted:**

- 3-channel vs 12-lead: Cannot extract pseudo-R' in V1 or pseudo-S in inferior leads (100% specificity features for AVNRT in clinical literature)
- Small dataset (n=46): LOO-CV used to maximise validation reliability. Results should be validated on a larger independent cohort.
- RP interval detection: At 200+ BPM, NeuroKit2 may detect the pre-excitation P of the next beat rather than the retrograde P of the current beat. RP values (~148ms AVNRT, ~182ms AVRT) exceed the clinical threshold (<90ms) suggesting this limitation applies.